

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Semester : V

Branch : Electronics and Telecommunication Engineering / Electronics and Communication Engineering

Subject Code & Name: BTETC503 - Analog Communication

Max Marks: 60

Date: 05/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Define Modulation. Explain need of modulation in detail.	CO01	6
B) State sampling theorem. Explain sampling process in detail. A message signal made of multiple frequency components: a single tone sine wave of 850 kHz and sound of frequency components with lowest value of 500 Hz and highest value of 2700 Hz. Calculate the minimum sampling frequency to sense the information present in this signal according to sampling theorem.	CO01	6
C) Explain following terms: (a) simplex system, (b) duplex system, (c) point to point communication, (d) broadcast communication, (e) TDM, (f) FDM.	CO01	6
Q.2 Solve Any Two of the following.		12
A) Explain generation of SSB signal using (i) the filter method and (ii) the phase shift method.	CO02	6
B) Explain Vestigial Sideband (VSB) modulation techniques along with its advantages and disadvantages.	CO02	6
C) A 750 W carrier is amplitude modulated to the depth of 75%. Calculate the total power in case of (i) DSB-SC, (ii) SSB-SC, (iii) ISB and (iv) VSB (with 25% of other sideband). Also find power saving compared to DSB-FC for each case.	CO02	6

Q. 3 Solve Any Two of the following.		12
A) Explain frequency modulation. Also find mathematical expression for (i) maximum frequency deviation, (ii) instantaneous value of FM voltage.	CO05	6
B) State the difference between AM, FM and PM.	CO05	6
C) In a PM system, when the audio frequency (AF) is 550 Hz, and the AF voltage is 3.2V, the deviation is 3.6 kHz. If the AF voltage is now increased to 7.2 V, what is new deviation? If the AF voltage is further raised by 7V while the AF is dropped to 175 Hz, what is the deviation? Find modulation index in each case.	CO05	6
Q.4 Solve Any Two of the following.		12
A) Enlist different types of radio receiver. Explain any one in detail with the help of block diagram.	CO03	6
B) Briefly explain any four performance characteristic of a radio receiver.	CO03, 04	6
C) Explain double spotting with its cause and effect. Explain how it can be reduced.	CO04	6
Q. 5 Solve Any Two of the following.		12
A) Explain operation of ratio detector. Also explain how amplitude limiting can be achieved using ratio detector	CO03	6
B) Define noise. Explain its various types in communication system.	CO06	6
C) a) Define (i) signal to noise ratio and (ii) noise figure. b) A receiver connected to antenna whose resistance is 80Ω has an equivalent noise resistance of 45Ω . Calculate the receiver's noise figure in decibels and its equivalent noise temperature.	CO06	6

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